

Ainekko MXM Board

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Modified 03/16/2026

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Voltage Rail Definitions and Ranges

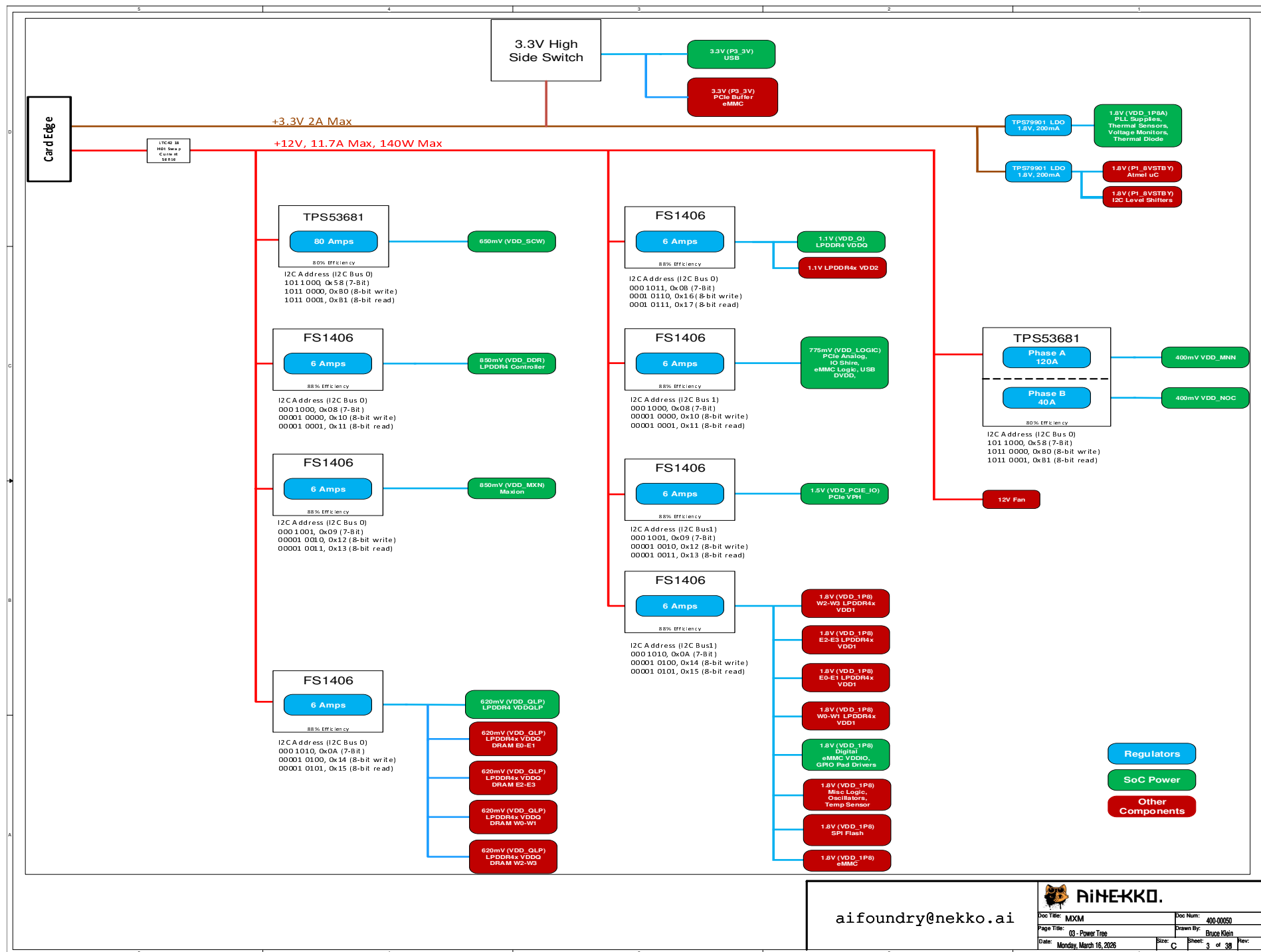
Voltage Rail	Boot Voltage	PMIC Supported Range	Where Used
VDD_MNN	500mV	305mV - 750mV	Minions and some Neighborhood
VDD_NOC	430mV	305mV - 750mV	NOCs
VDD_SCW	750mV	550mV - 900mV	SCW logic including L2/L3 SRAM, 4x4 crossbar, Neighborhood 32KB shared L1 caches/logic, level shifters, LVPLLs, Sprinkler
VDD_MXN	850mV	650mV - 885mV	4 Maxions and their caches
VDD_DDR	800mV	700mV - 950mV	DRAM controllers (synopsys IP for 1066MHz) in MemShire
VDD_LOGIC	775mV	700mV - 900mV	Logic in IO shires, GPIO, PShire, fuse blocks, DFT
VDD_Q	1.1V	1.06V - 1.17V	DRAM internals, SoC/DRAM PHYs
VDD_QLP	640mV	530mV - 630mV	IO voltage to DRAM and SoC
VDD_PCIE_IO	1.5V	1.4V - 1.8V	PCIe IO
P1_8VSTDBY	1.8V	1.8V	PMIC Controller
VDDA_1P8	1.8V	1.8V	Analog 1.8V
VDD_1P8	1.8V	1.8V	SoC GPIO voltages, random board logic
P3_3V	3.3V	3.3V	USB and random board components

Note: Boot Voltages based upon SoC FW 0.8.0, PMIC FW 0.7.0
Values may change. Consult the firmware release documents.

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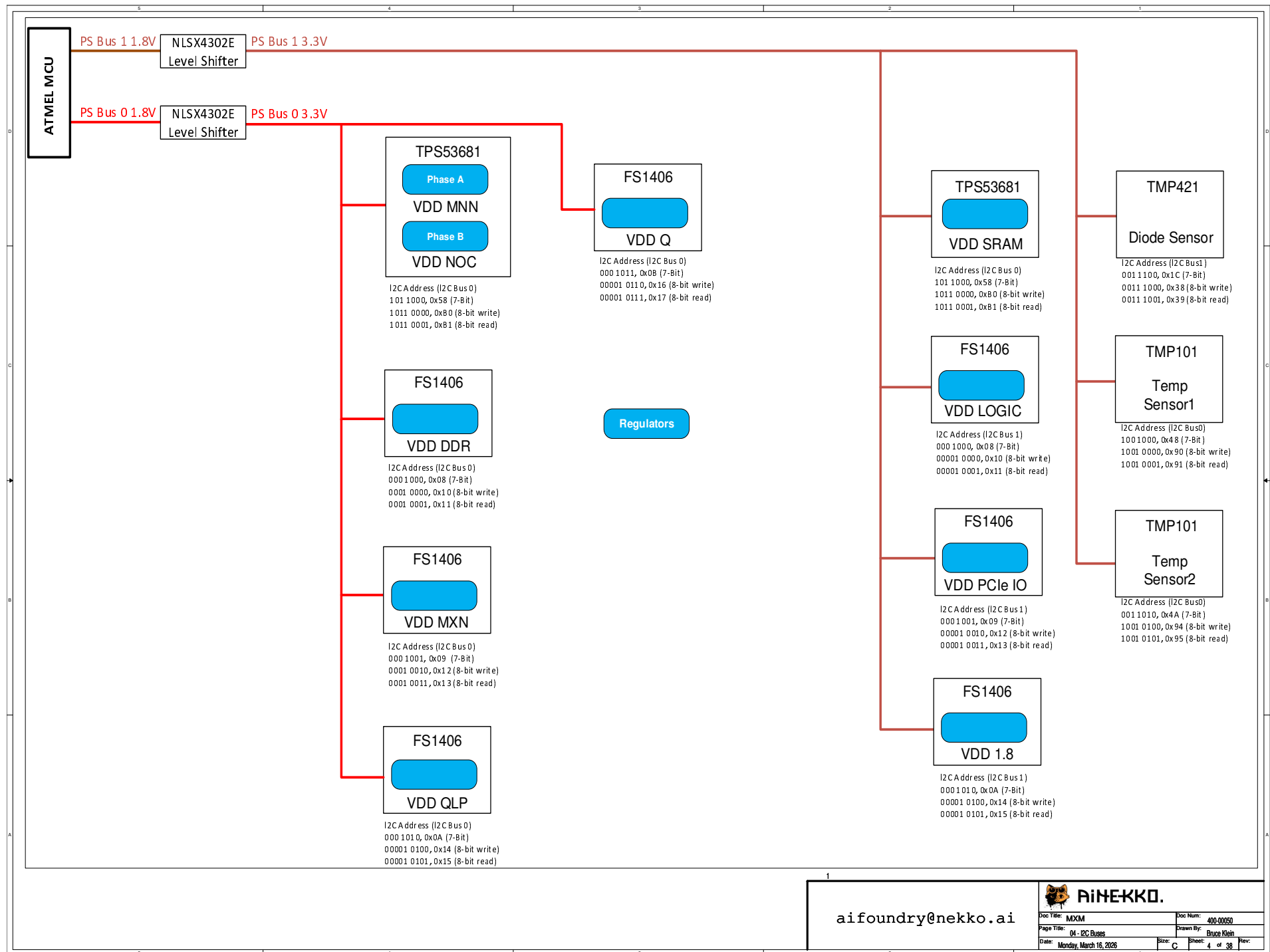
Doc Title: MXM	Doc Num: 400-00050
Page Title: 02 - Voltage Rail Definitions	Drawn By: Bruce Klein
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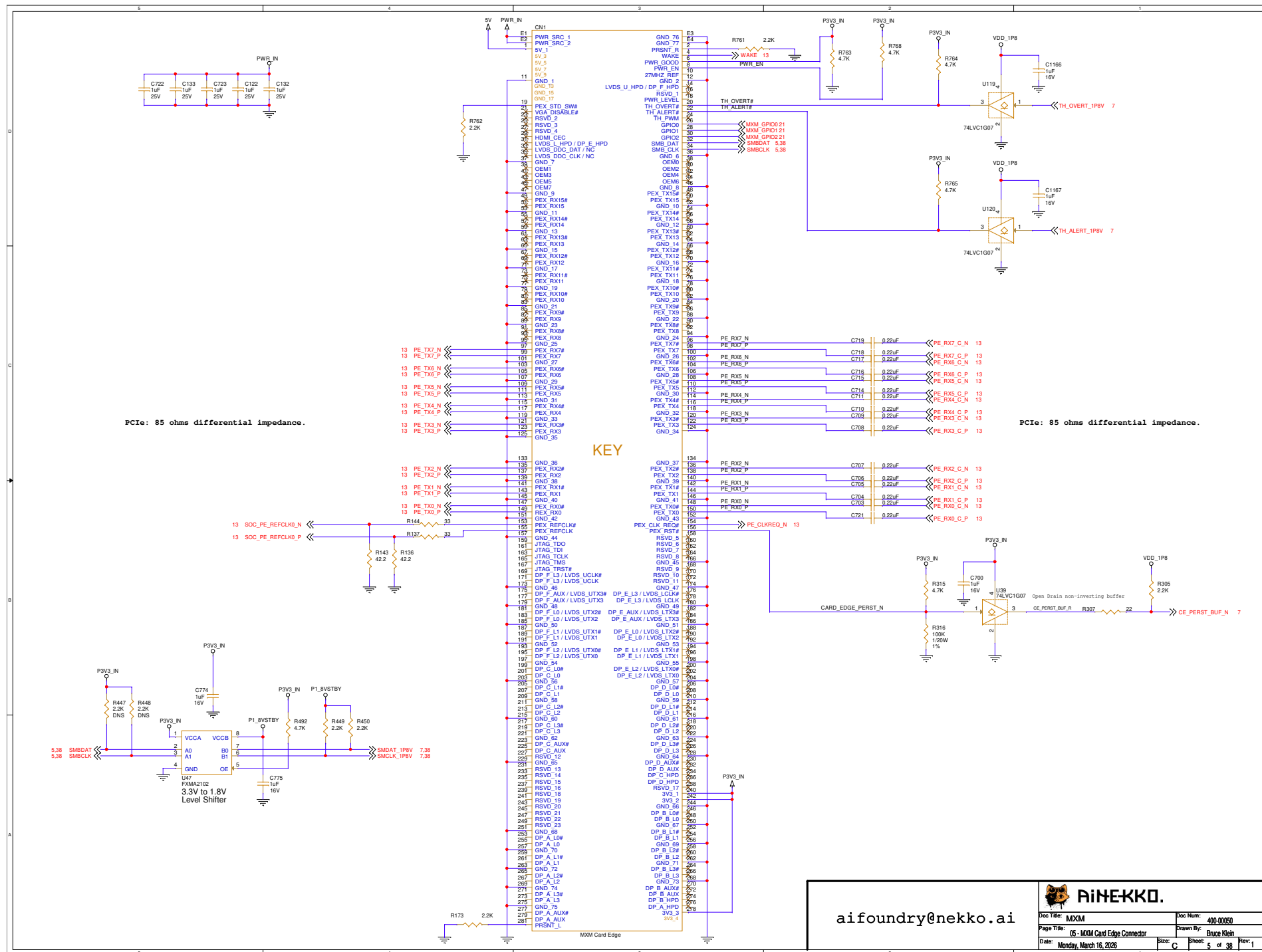
Doc Title: MXM	Doc Num: 400-00050
Page Title: 03 - Power Tree	Drawn By: Bruce Klein
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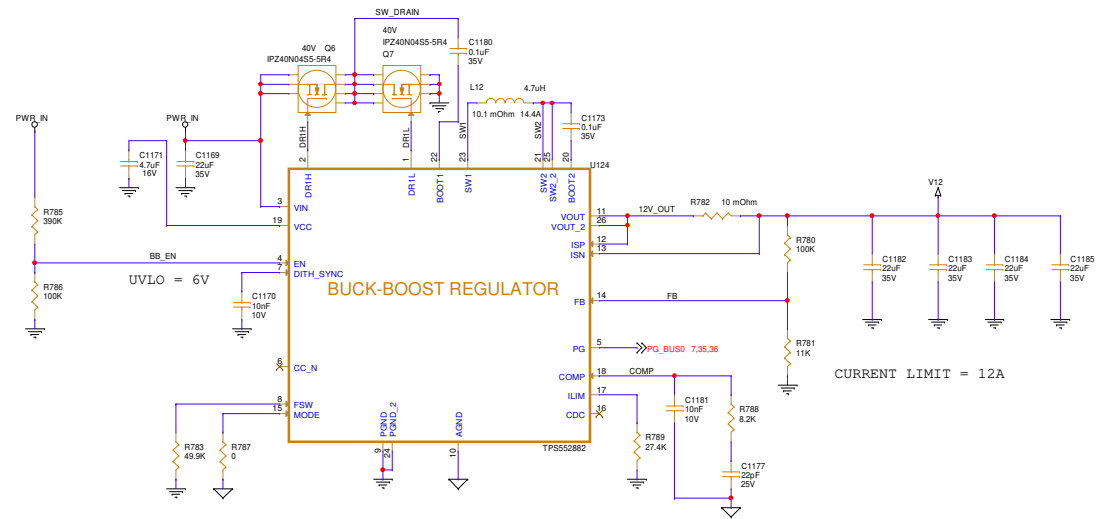
Doc Title: MXM	Doc Num: 400-00050
Page Title: 04 - I2C Buses	Drawn By: Bruce Klein
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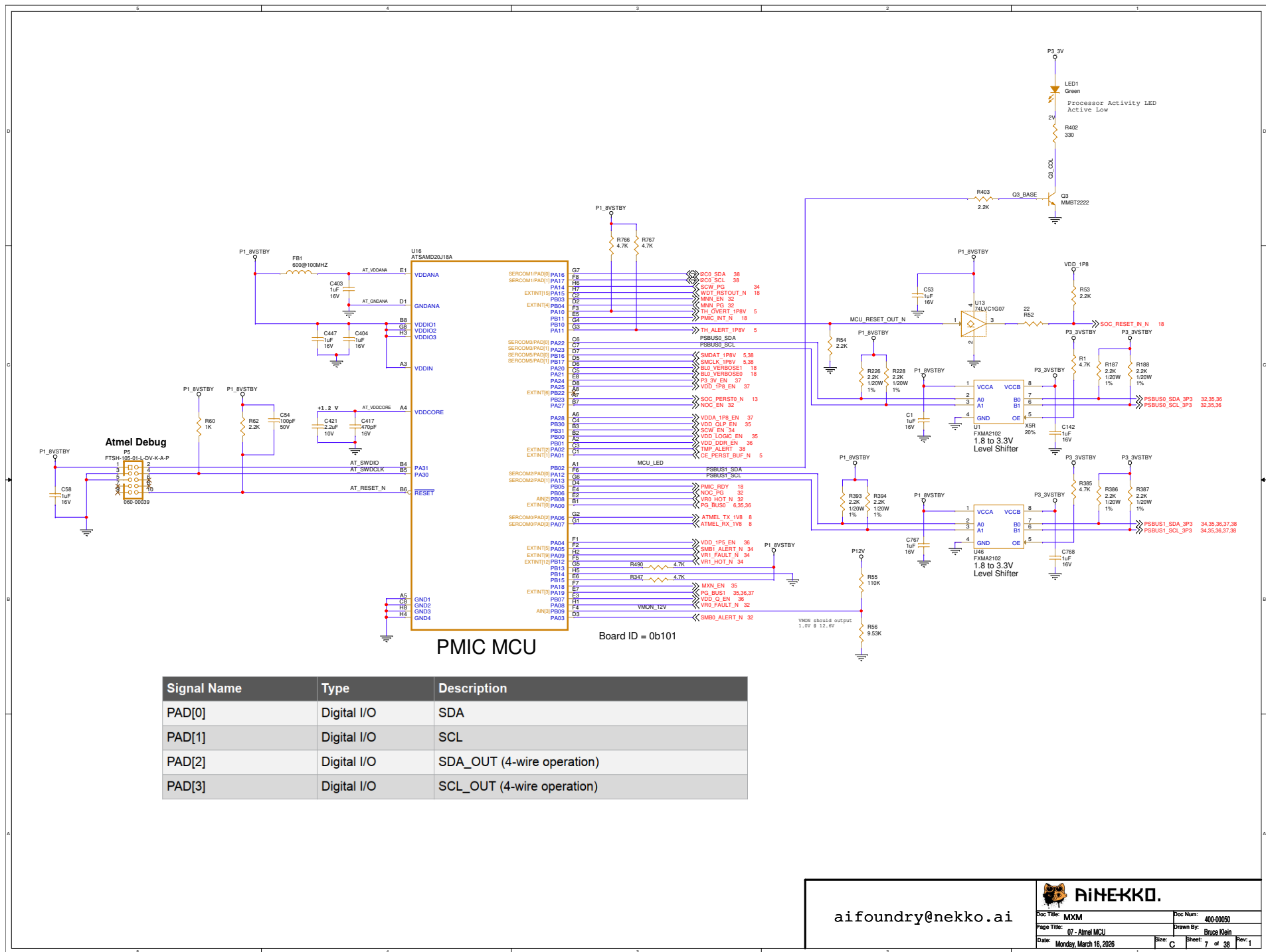
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Page Title: 05 - MXM Card Edge Connector		Drawn By: Bruce Klein	
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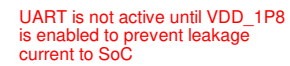
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Date: Monday, March 16, 2026	Size: C Sheet: 6 of 38 Rev: <Rev>

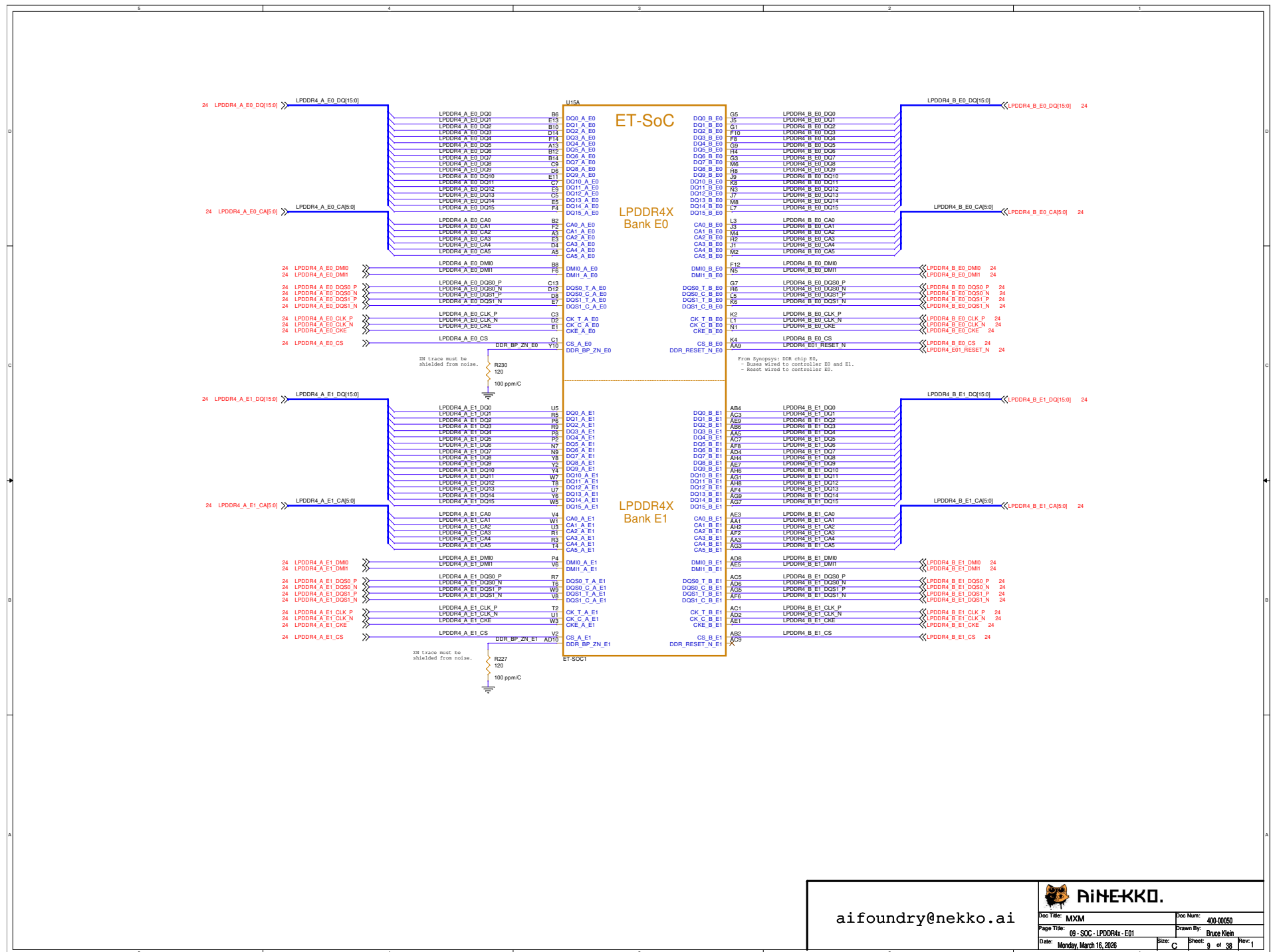


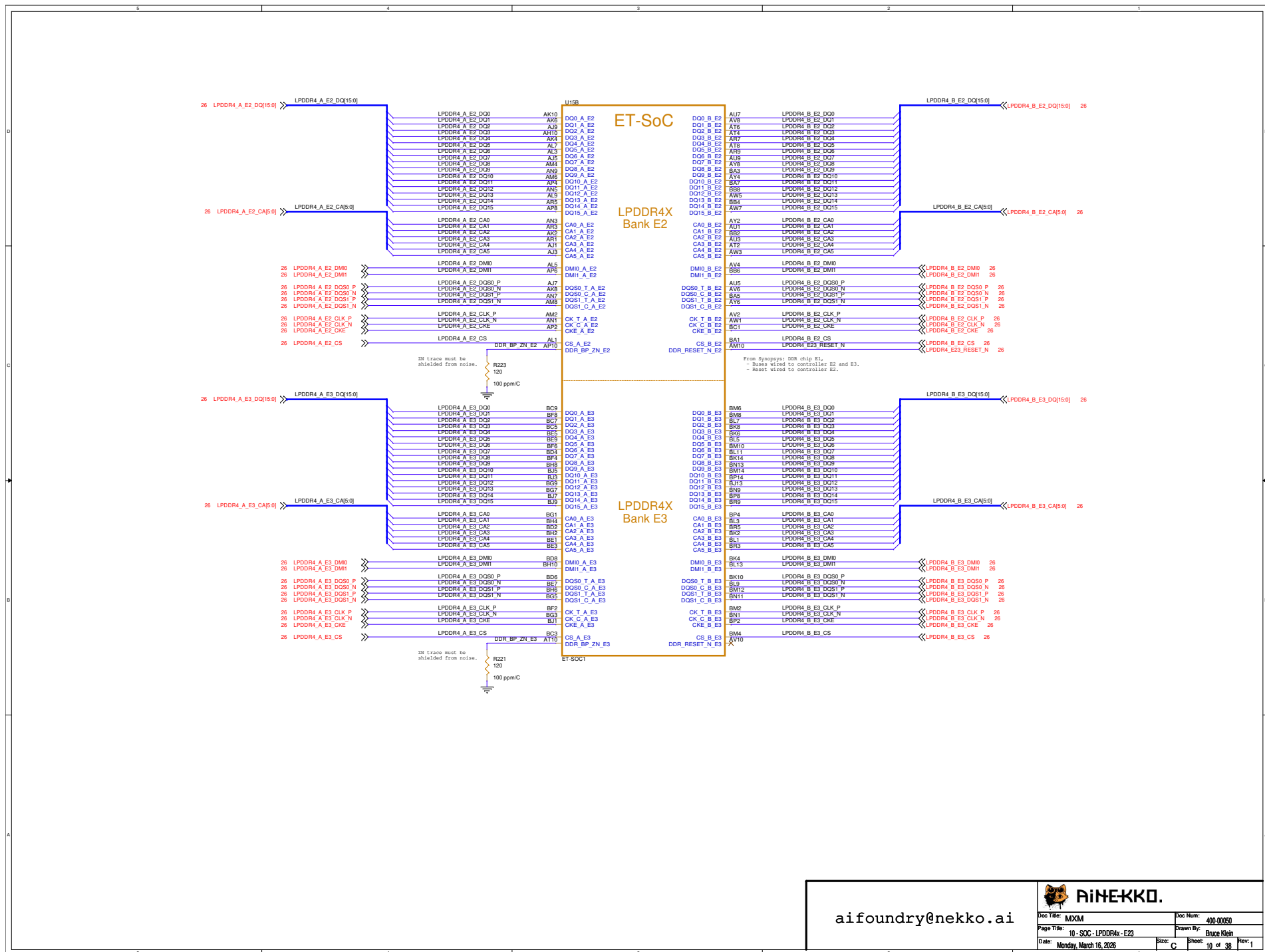
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Page Title: 07 - Atmel MCU	Drawn By: Bruce Klein
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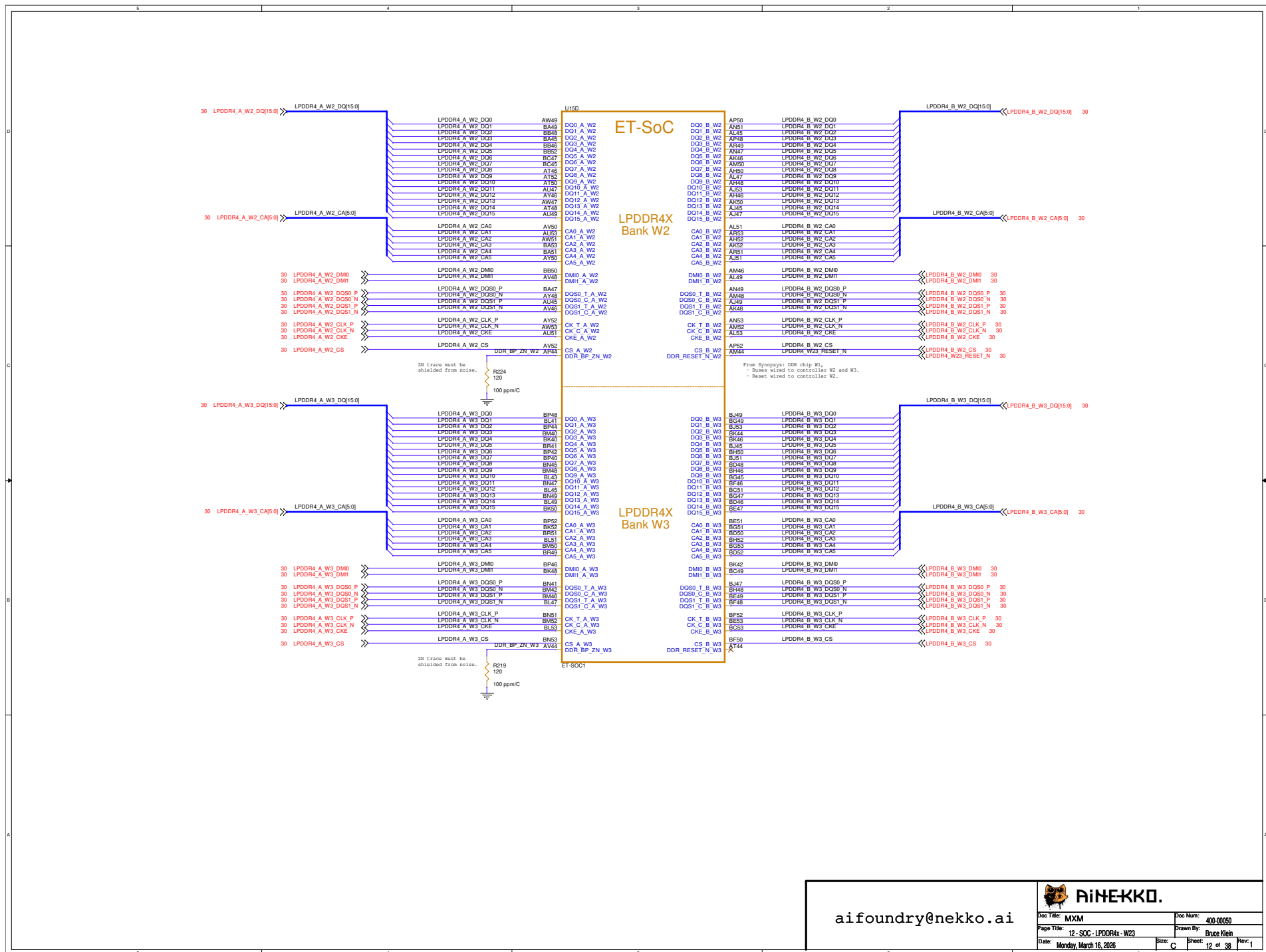


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Page Title: 10 - SOC - LPDDR4x - E23	Drawn By: Bruce Klein
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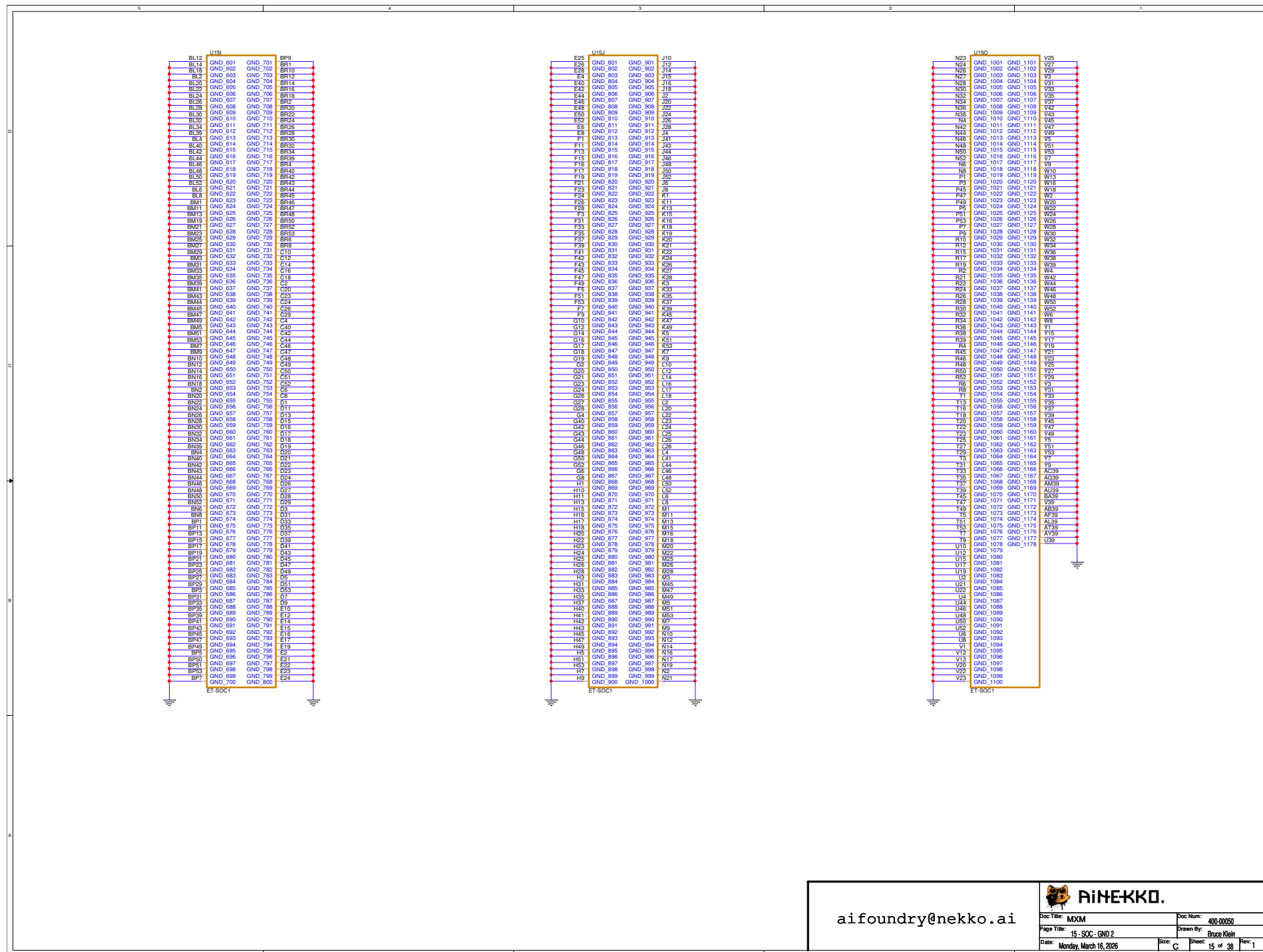
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Page Title: 12-SOC-LPDDR4x-W23	Drawn By: Bruce Klein
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U15F			
A1	GND 1	GND 101	AE29
A10	GND 2	GND 102	AE31
A12	GND 3	GND 103	AE33
A14	GND 4	GND 104	AE35
A16	GND 5	GND 105	AE37
A18	GND 6	GND 106	AE39
A2	GND 7	GND 107	AE41
A4	GND 8	GND 108	AE43
A6	GND 9	GND 109	AE45
A8	GND 10	GND 110	AE47
A10	GND 11	GND 111	AE49
A12	GND 12	GND 112	AE51
A14	GND 13	GND 113	AE53
A16	GND 14	GND 114	AE55
A18	GND 15	GND 115	AE57
A20	GND 16	GND 116	AE59
A22	GND 17	GND 117	AE61
A24	GND 18	GND 118	AE63
A26	GND 19	GND 119	AE65
A28	GND 20	GND 120	AE67
A30	GND 21	GND 121	AE69
A32	GND 22	GND 122	AE71
A34	GND 23	GND 123	AE73
A36	GND 24	GND 124	AE75
A38	GND 25	GND 125	AE77
A40	GND 26	GND 126	AE79
A42	GND 27	GND 127	AE81
A44	GND 28	GND 128	AE83
A46	GND 29	GND 129	AE85
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A54	GND 33	GND 133	AE93
A56	GND 34	GND 134	AE95
A58	GND 35	GND 135	AE97
A60	GND 36	GND 136	AE99
A62	GND 37	GND 137	AE101
A64	GND 38	GND 138	AE103
A66	GND 39	GND 139	AE105
A68	GND 40	GND 140	AE107
A70	GND 41	GND 141	AE109
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A74	GND 43	GND 143	AE113
A76	GND 44	GND 144	AE115
A78	GND 45	GND 145	AE117
A80	GND 46	GND 146	AE119
A82	GND 47	GND 147	AE121
A84	GND 48	GND 148	AE123
A86	GND 49	GND 149	AE125
A88	GND 50	GND 150	AE127
A90	GND 51	GND 151	AE129
A92	GND 52	GND 152	AE131
A94	GND 53	GND 153	AE133
A96	GND 54	GND 154	AE135
A98	GND 55	GND 155	AE137
A100	GND 56	GND 156	AE139
A102	GND 57	GND 157	AE141
A104	GND 58	GND 158	AE143
A106	GND 59	GND 159	AE145
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A110	GND 61	GND 161	AE149
A112	GND 62	GND 162	AE151
A114	GND 63	GND 163	AE153
A116	GND 64	GND 164	AE155
A118	GND 65	GND 165	AE157
A120	GND 66	GND 166	AE159
A122	GND 67	GND 167	AE161
A124	GND 68	GND 168	AE163
A126	GND 69	GND 169	AE165
A128	GND 70	GND 170	AE167
A130	GND 71	GND 171	AE169
A132	GND 72	GND 172	AE171
A134	GND 73	GND 173	AE173
A136	GND 74	GND 174	AE175
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A146	GND 79	GND 179	AE185
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A150	GND 81	GND 181	AE189
A152	GND 82	GND 182	AE191
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A160	GND 86	GND 186	AE199
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A170	GND 91	GND 191	AE209
A172	GND 92	GND 192	AE211
A174	GND 93	GND 193	AE213
A176	GND 94	GND 194	AE215
A178	GND 95	GND 195	AE217
A180	GND 96	GND 196	AE219
A182	GND 97	GND 197	AE221
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A186	GND 99	GND 199	AE225
A188	GND 100	GND 200	AE227

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AK51	GND 202	GND 302	AR8
AK53	GND 203	GND 303	AT1
AK55	GND 204	GND 304	AT3
AK57	GND 205	GND 305	AT5
AK59	GND 206	GND 306	AT7
AK61	GND 207	GND 307	AT9
AK63	GND 208	GND 308	AT11
AK65	GND 209	GND 309	AT13
AK67	GND 210	GND 310	AT15
AK69	GND 211	GND 311	AT17
AK71	GND 212	GND 312	AT19
AK73	GND 213	GND 313	AT21
AK75	GND 214	GND 314	AT23
AK77	GND 215	GND 315	AT25
AK79	GND 216	GND 316	AT27
AK81	GND 217	GND 317	AT29
AK83	GND 218	GND 318	AT31
AK85	GND 219	GND 319	AT33
AK87	GND 220	GND 320	AT35
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AK93	GND 223	GND 323	AT41
AK95	GND 224	GND 324	AT43
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AK99	GND 226	GND 326	AT47
AK101	GND 227	GND 327	AT49
AK103	GND 228	GND 328	AT51
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AK121	GND 237	GND 337	AT69
AK123	GND 238	GND 338	AT71
AK125	GND 239	GND 339	AT73
AK127	GND 240	GND 340	AT75
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AK131	GND 242	GND 342	AT79
AK133	GND 243	GND 343	AT81
AK135	GND 244	GND 344	AT83
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AK139	GND 246	GND 346	AT87
AK141	GND 247	GND 347	AT89
AK143	GND 248	GND 348	AT91
AK145	GND 249	GND 349	AT93
AK147	GND 250	GND 350	AT95
AK149	GND 251	GND 351	AT97
AK151	GND 252	GND 352	AT99
AK153	GND 253	GND 353	AT101
AK155	GND 254	GND 354	AT103
AK157	GND 255	GND 355	AT105
AK159	GND 256	GND 356	AT107
AK161	GND 257	GND 357	AT109
AK163	GND 258	GND 358	AT111
AK165	GND 259	GND 359	AT113
AK167	GND 260	GND 360	AT115
AK169	GND 261	GND 361	AT117
AK171	GND 262	GND 362	AT119
AK173	GND 263	GND 363	AT121
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AK227	GND 290	GND 390	AT175
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AK245	GND 299	GND 399	AT193
AK247	GND 300	GND 400	AT195

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B27	GND 404	GND 504	BF11
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B31	GND 406	GND 506	BF15
B33	GND 407	GND 507	BF17
B35	GND 408	GND 508	BF19
B37	GND 409	GND 509	BF21
B39	GND 410	GND 510	BF23
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B45	GND 413	GND 513	BF29
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B213	GND 497	GND 597	BF197
B215	GND 498	GND 598	BF199
B217	GND 499	GND 599	BF201
B219	GND 500	GND 600	BF203
BF51			

ET-5001



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BC15 NC.1 BD44
BC16 NC.2 NC.101 BE12
BC17 NC.3 NC.102 BE15
BC18 NC.4 NC.103 BE17
BD19 NC.5 NC.104 BE20
BD20 NC.6 NC.105 BE21
BE19 NC.7 NC.106 BE22
BE20 NC.8 NC.107 BE23
BE21 NC.9 NC.108 BE24
BE22 NC.10 NC.109 BE25
BE23 NC.11 NC.110 BE26
BE24 NC.12 NC.111 BE27
BE25 NC.13 NC.112 BE28
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BE76 NC.64 NC.163 BE79
BE77 NC.65 NC.164 BE80
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BE82 NC.70 NC.169 BE85
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BE05 NC.93 NC.192 BE08
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BE10 NC.98 NC.197 BE13
BE11 NC.99 NC.198 BE14
BE12 NC.100 NC.199 BE15
ET-SOC1

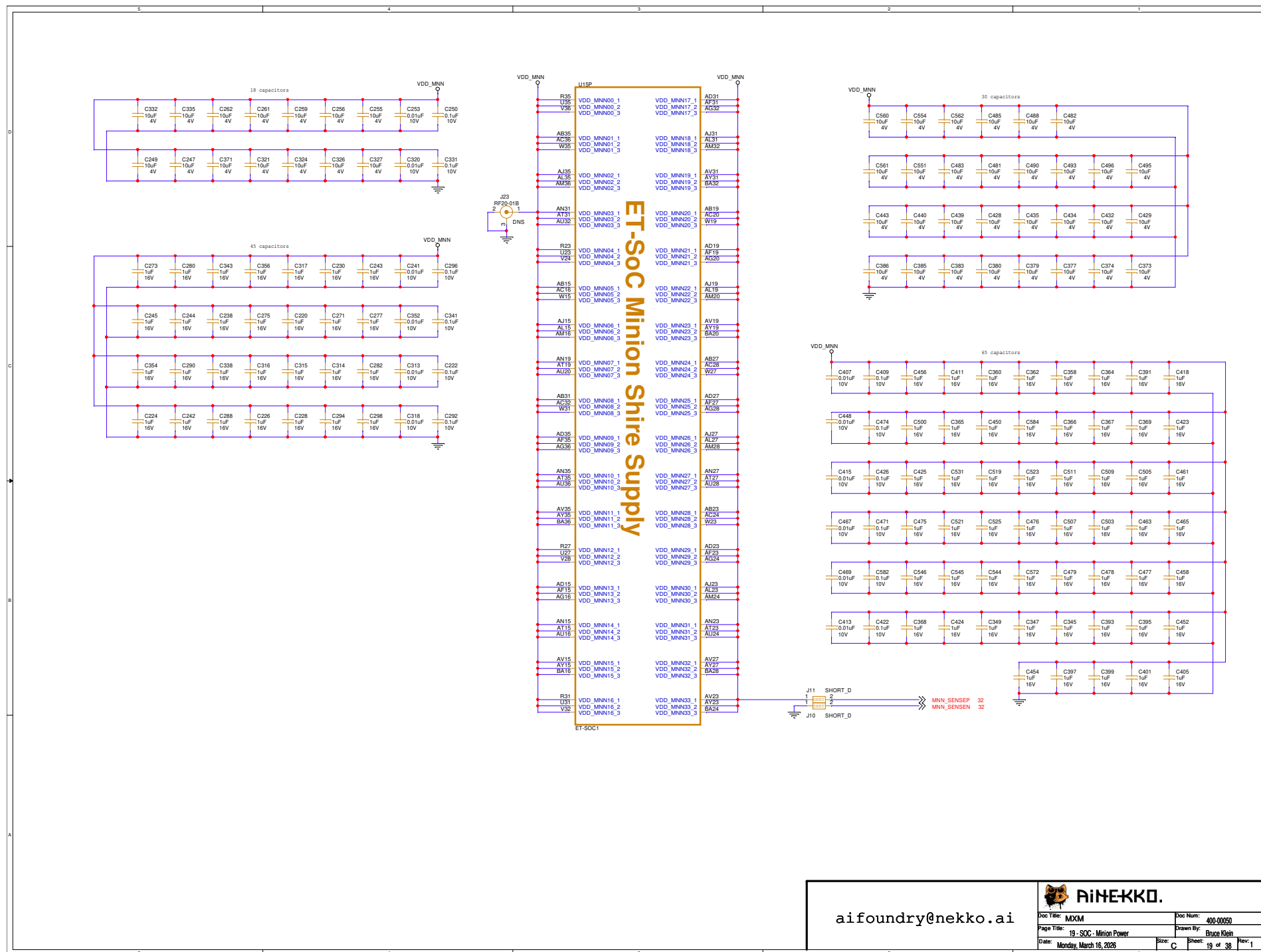
U19M
BM26 NC.201
BM27 NC.202
BM28 NC.203
BM29 NC.204
BM30 NC.205
BM31 NC.206
BM32 NC.207
BM33 NC.208
BM34 NC.209
BM35 NC.210
BM36 NC.211
BM37 NC.212
BM38 NC.213
BM39 NC.214
BM40 NC.215
BM41 NC.216
BM42 NC.217
BM43 NC.218
BM44 NC.219
BM45 NC.220
BM46 NC.221
BM47 NC.222
BM48 NC.223
BM49 NC.224
BM50 NC.225
BM51 NC.226
BM52 NC.227
BM53 NC.228
BM54 NC.229
BM55 NC.230
BM56 NC.231
BM57 NC.232
BM58 NC.233
BM59 NC.234
BM60 NC.235
BM61 NC.236
BM62 NC.237
BM63 NC.238
BM64 NC.239
BM65 NC.240
BM66 NC.241
BM67 NC.242
ET-SOC1

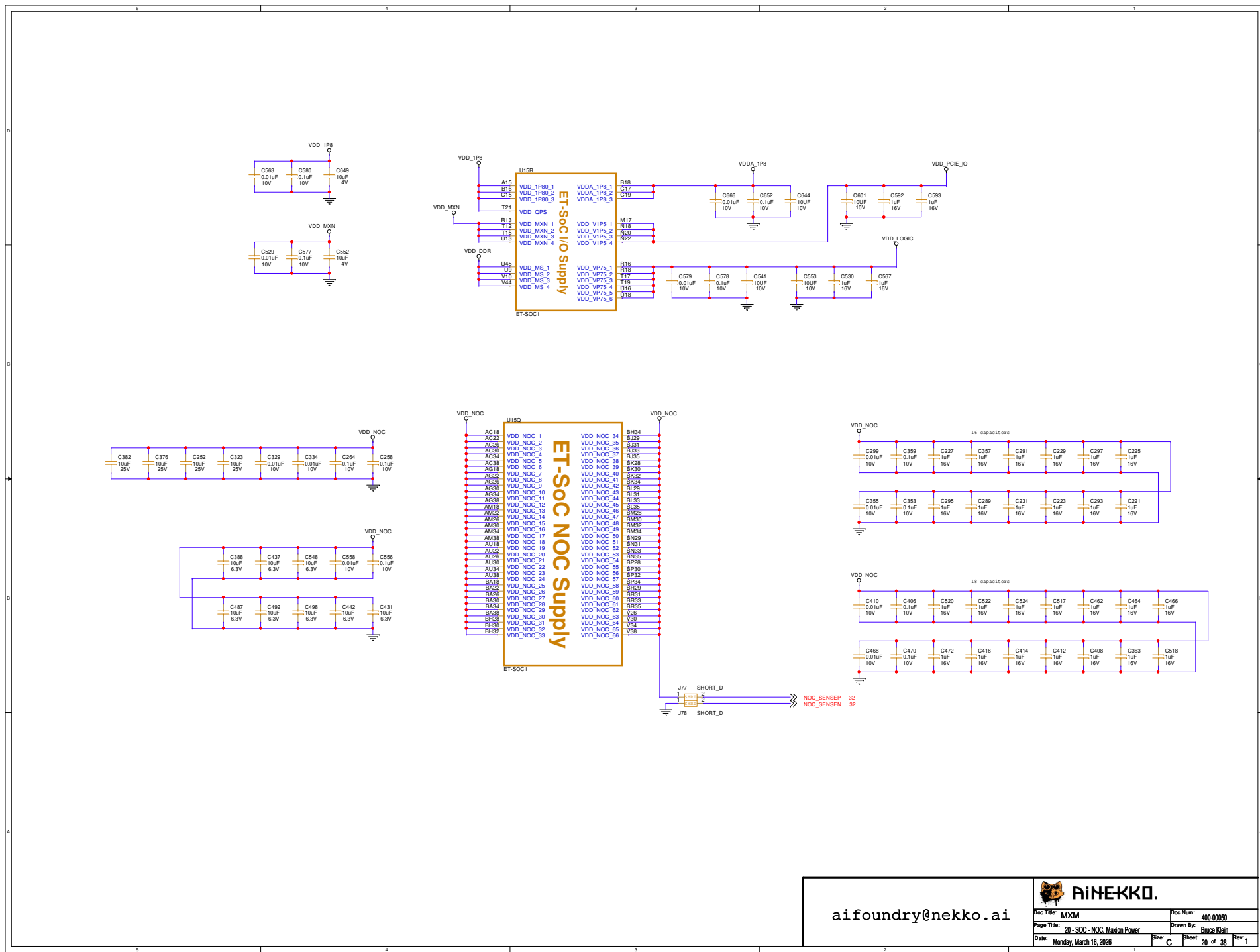
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Doc Title: MXM	Doc Num: 400-00050
Page Title: 17 - SOC - NC Pins	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 17 of 38 Rev: 1

SOC - Pin Strap Inputs



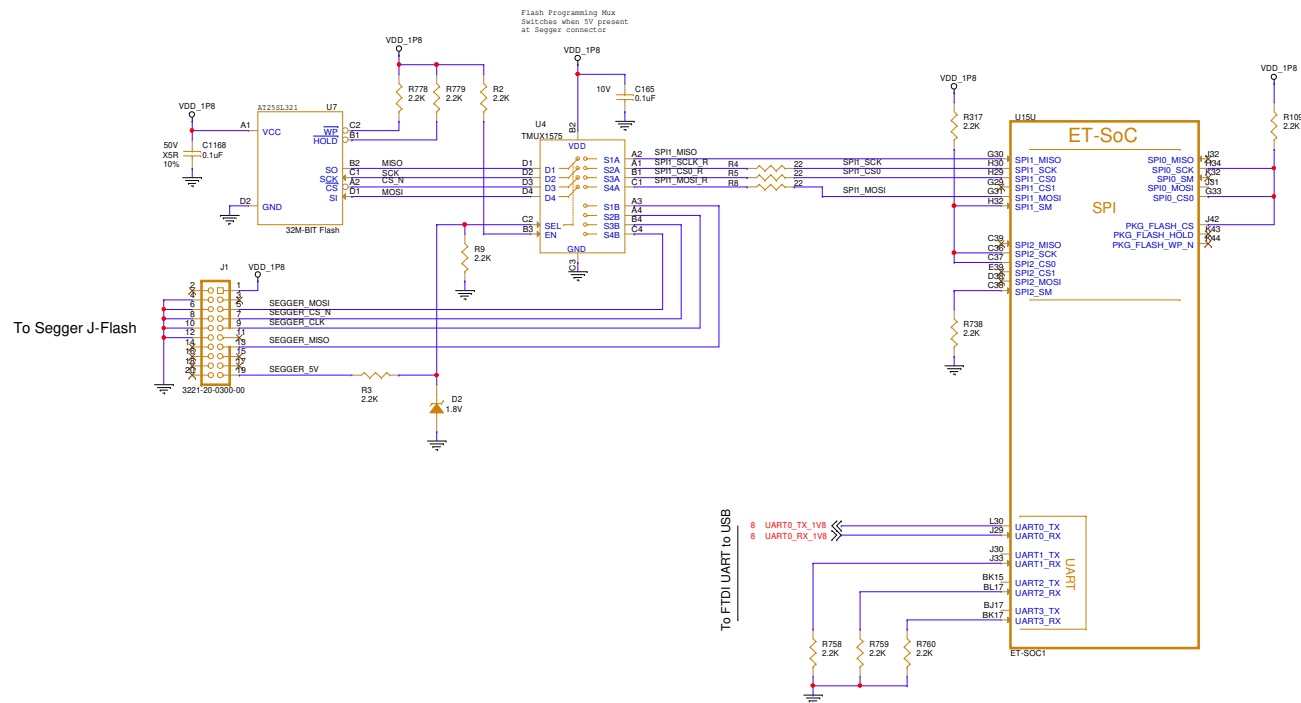


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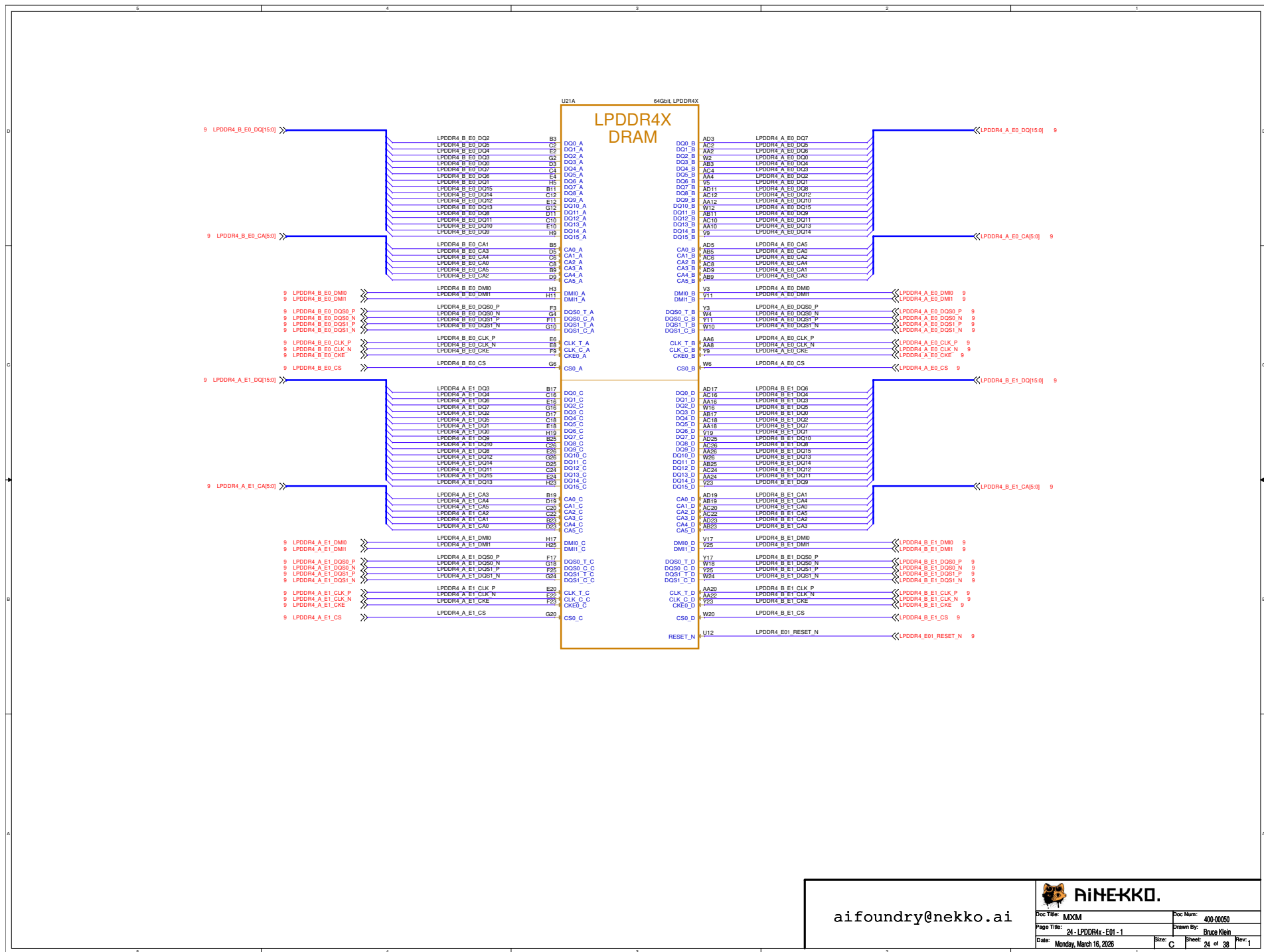
Doc Title: MXM	Doc Num: 400-00050
Page Title: 20 - SOC - NOC, Maxion Power	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 20 of 38 Rev: 1



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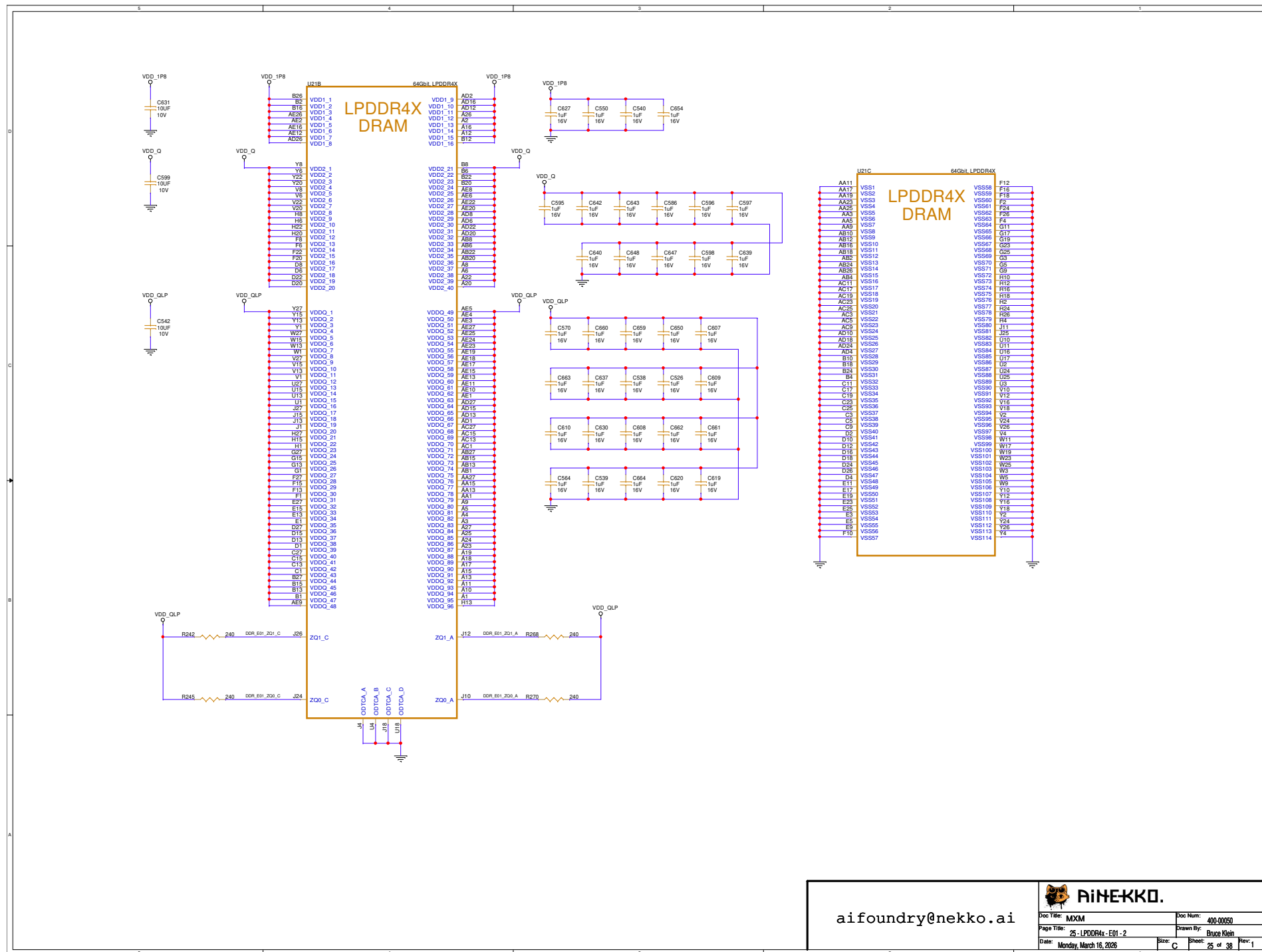
Doc Title: MXM	Doc Num: 400-00050
Page Title: 22-SOC-SPI, UART	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 22 of 38 Rev: 1

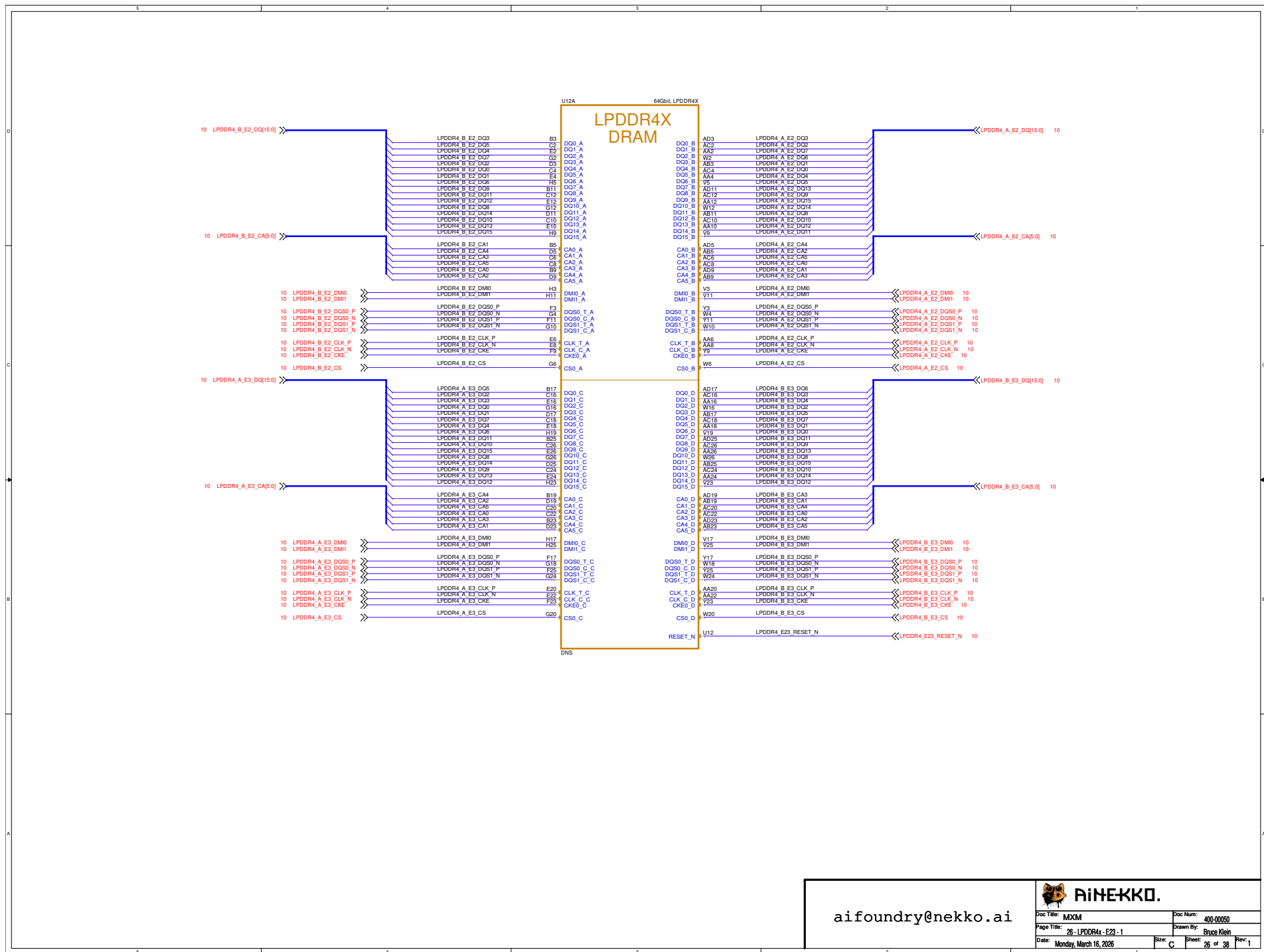


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Doc Title: MXM	Doc Num: 400-00050
Page Title: 24 - LPDDR4x - E01 - 1	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 24 of 38 Rev: 1

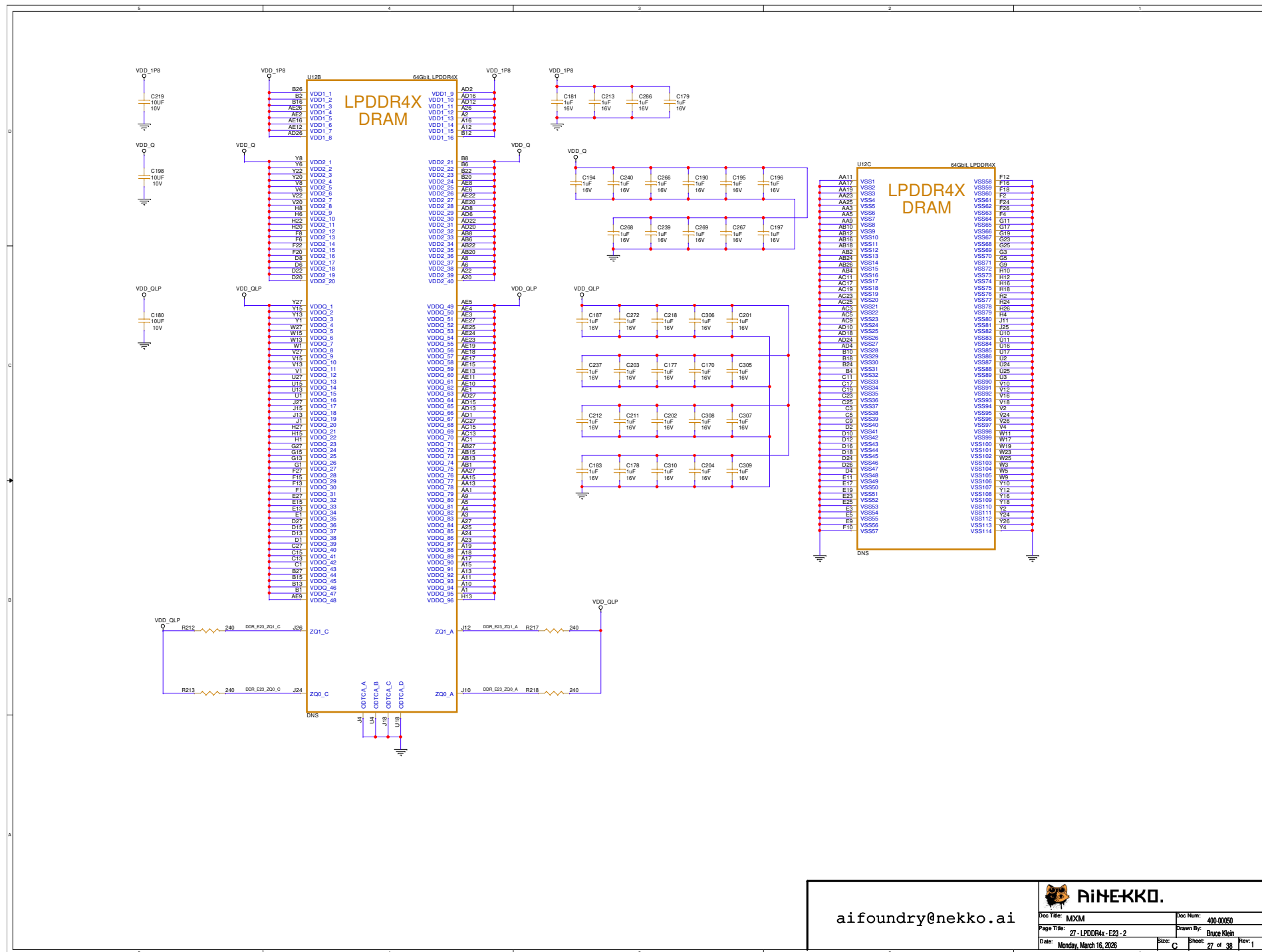


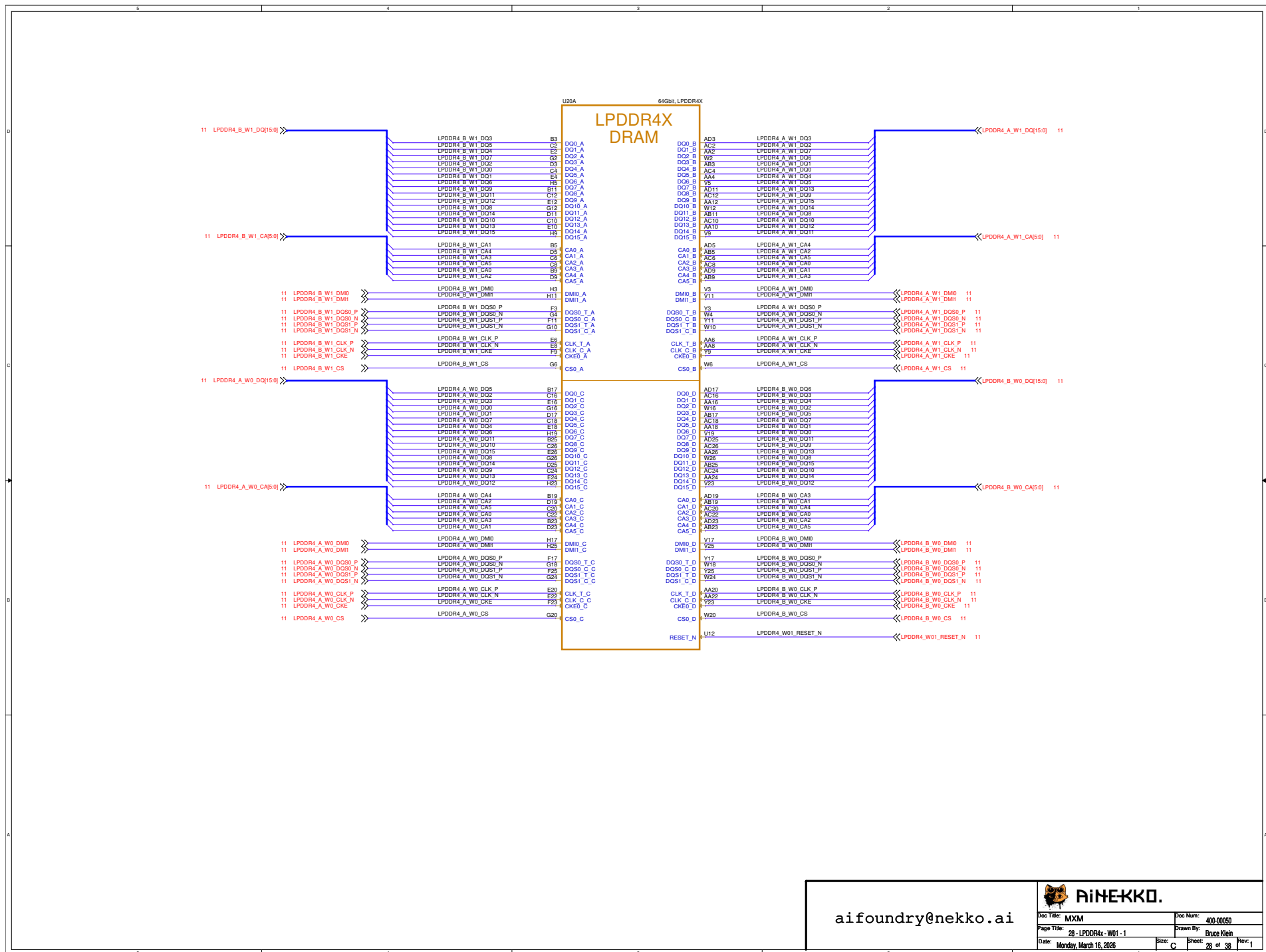


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Doc Title: MXM	Doc Num: 400-00050
Page Title: 26 - LPDDR4x - E23 - 1	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 26 of 38 Rev: 1



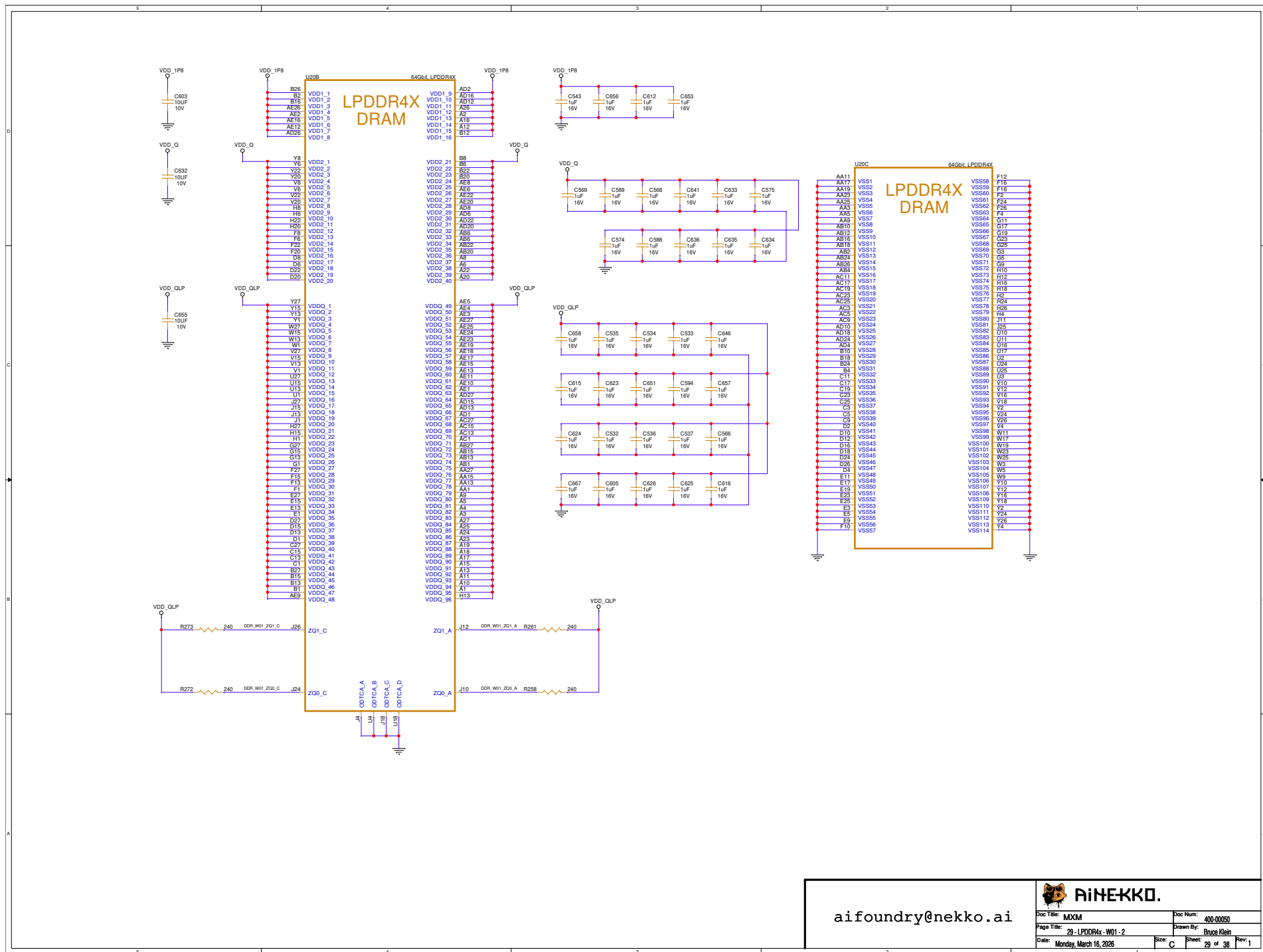


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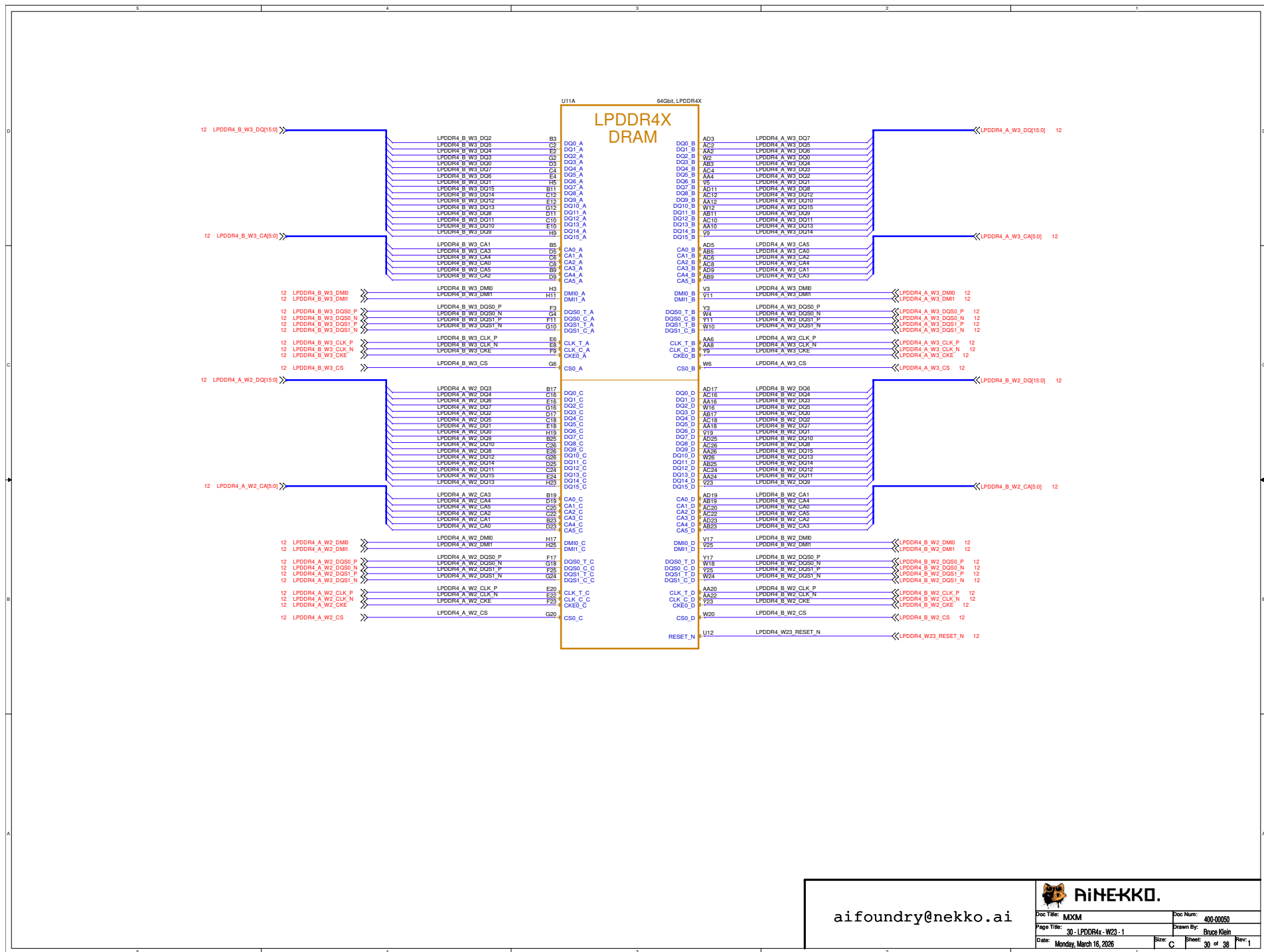
Doc Title: MXM	Doc Num: 400-00050
Page Title: 28 - LPDDR4x - W01 - 1	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 28 of 38 Rev: 1



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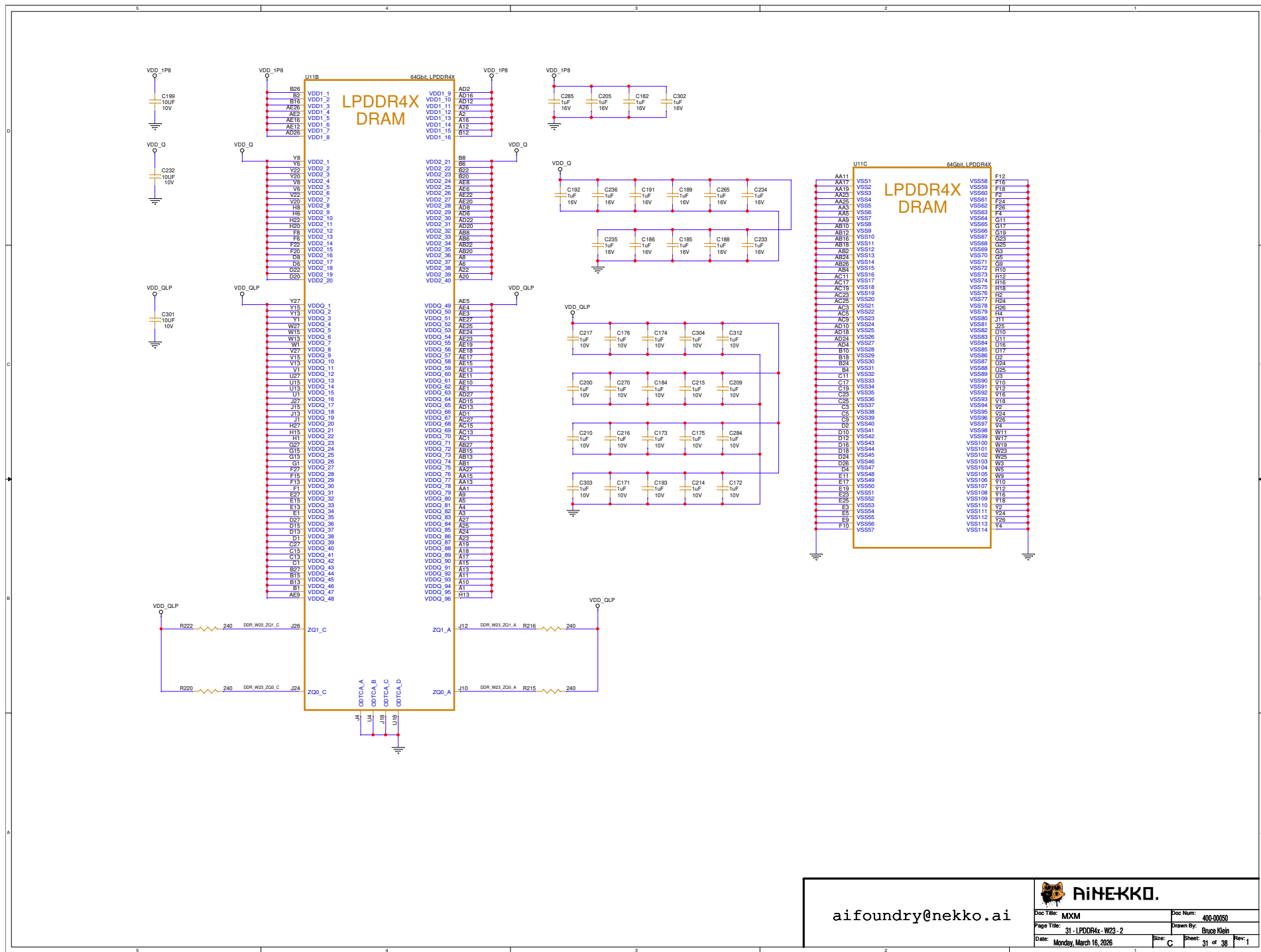
Doc Title: MXM	Doc Num: 400-00050
Page Title: 29 - LPDDR4x - W01 - 2	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 29 of 38 Rev: 1

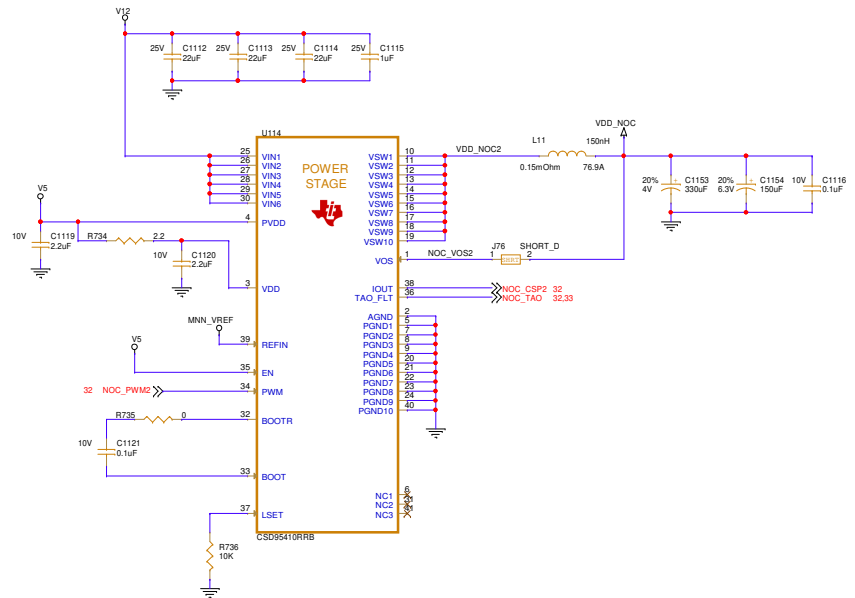
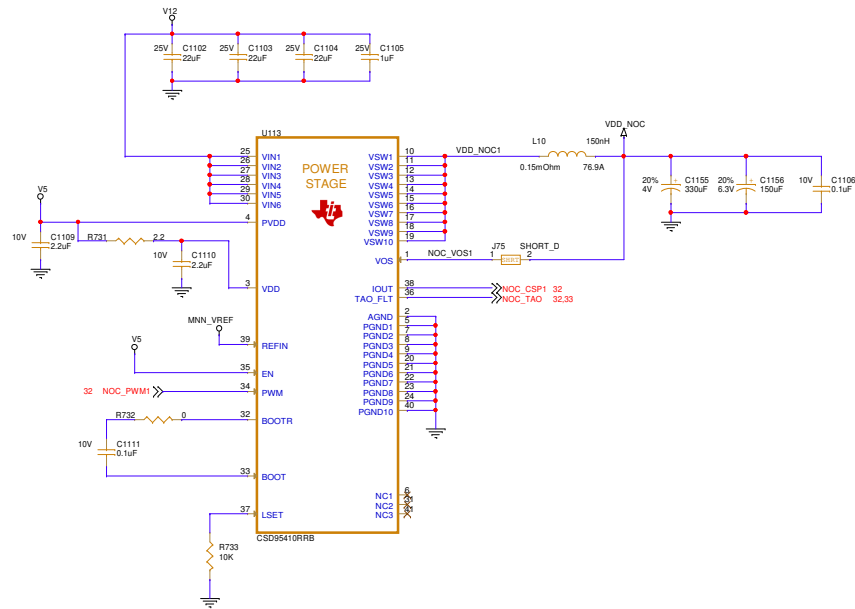


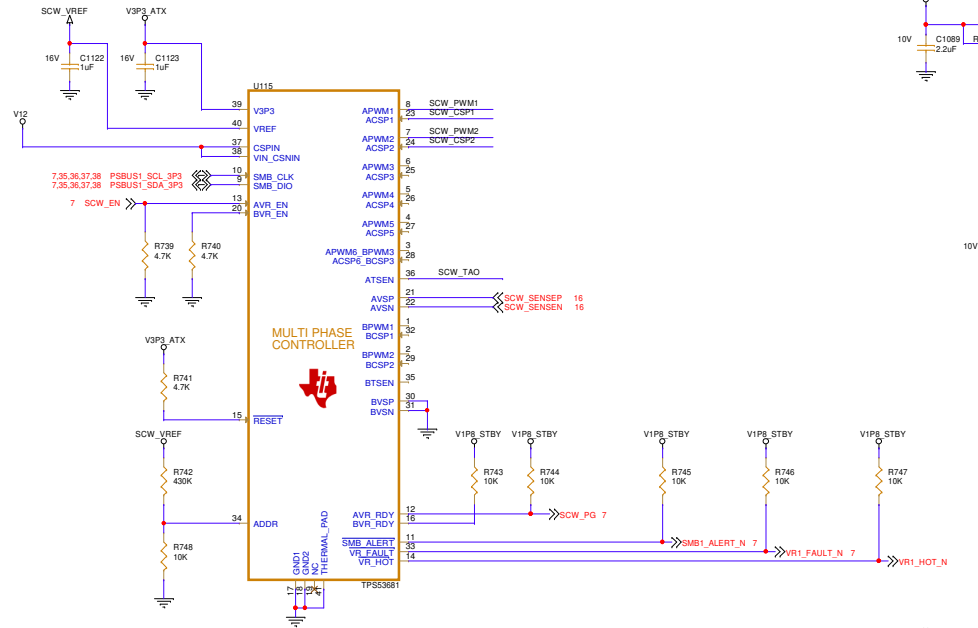
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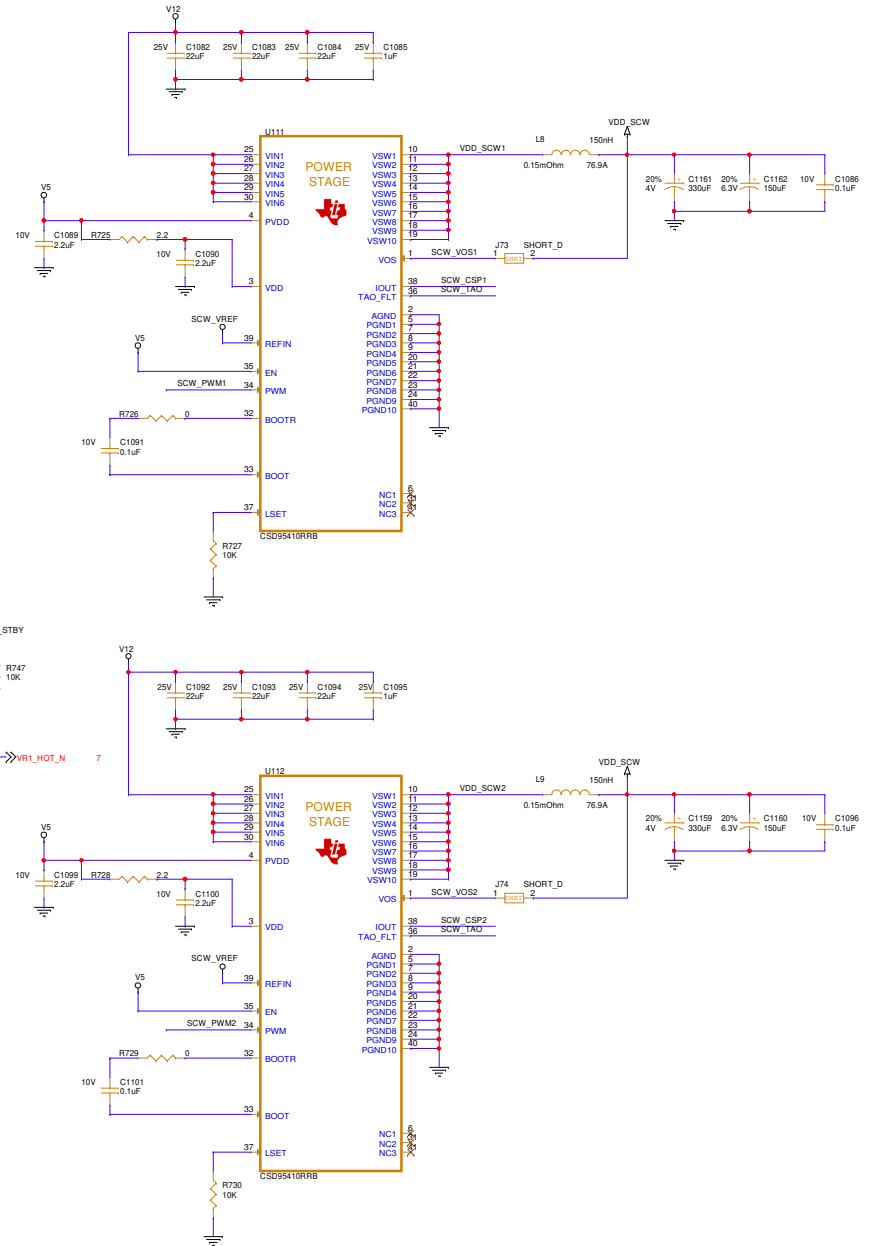
Doc Title: MXM	Doc Num: 400-00050
Page Title: 30 - LPDDR4x - W23 - 1	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 30 of 38 Rev: 1







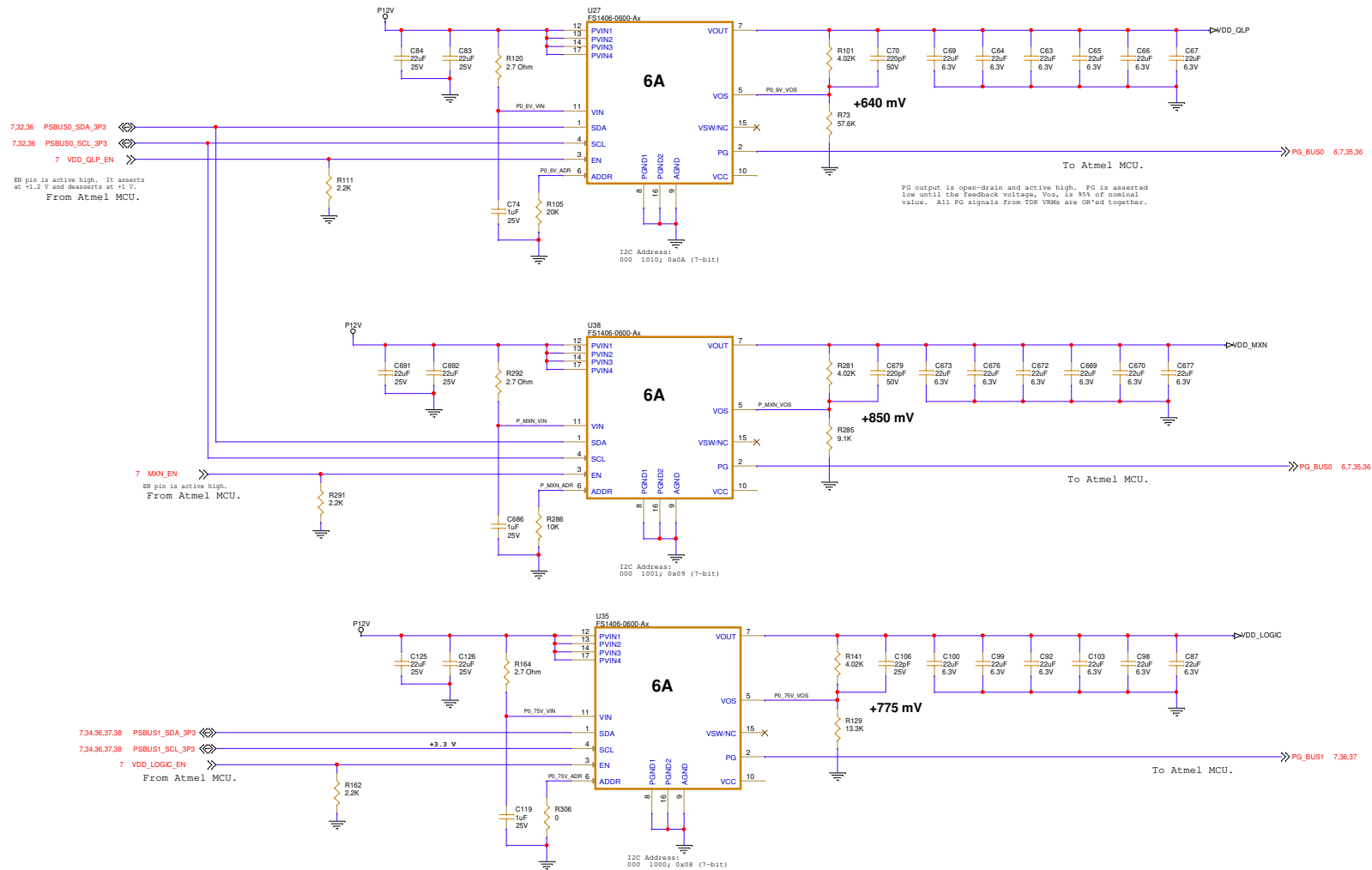
I2C Address
 0x58, 0b101_1000 (7-Bit)
 0xB0, 0b1011_0000 (8-Bit Write)
 0xB1, 0b1011_0001 (8-Bit Read)



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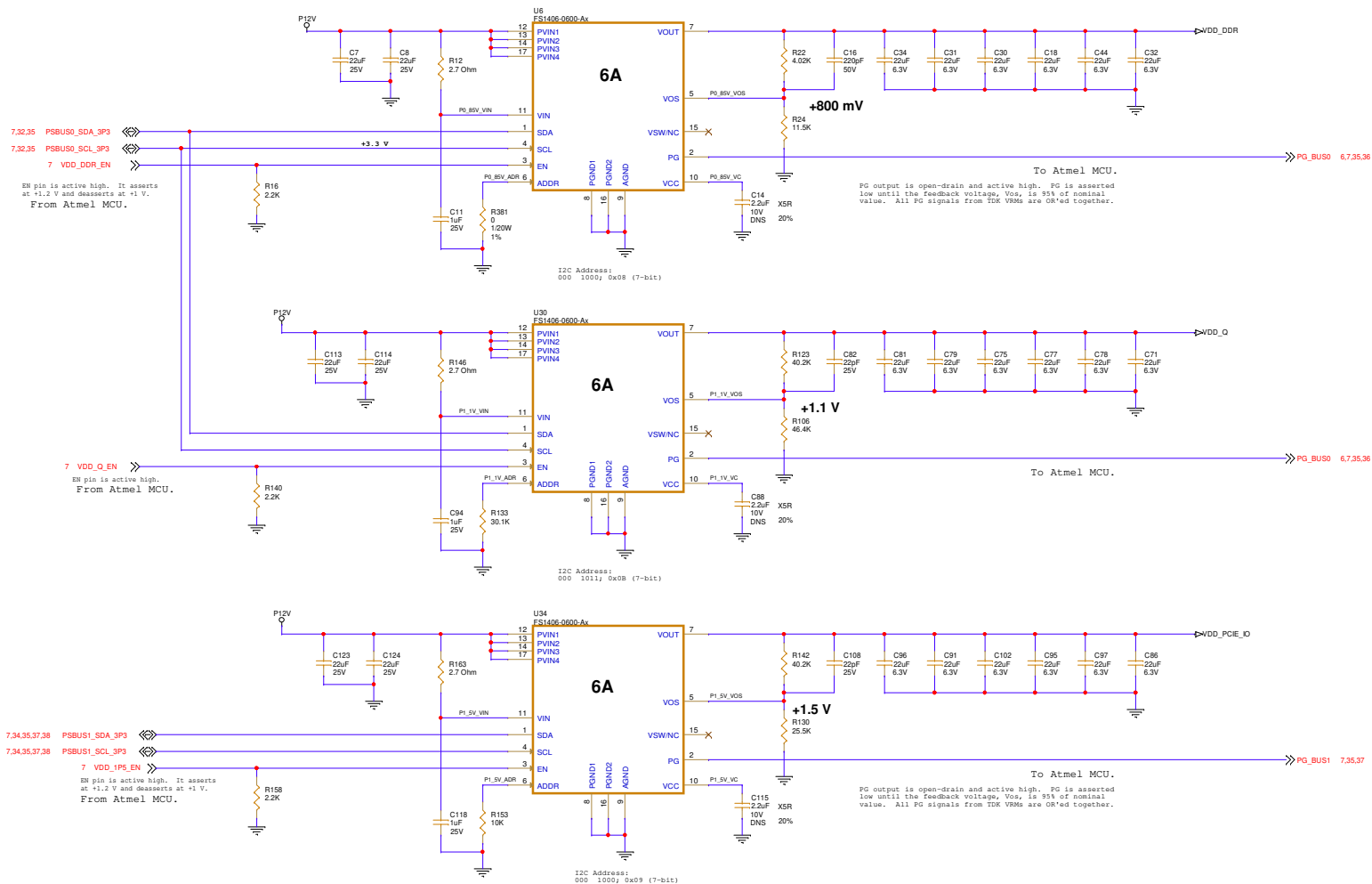
Doc Title: MXM	Doc Num: 400-00050
Page Title: 34 - PWR SCW	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 34 of 38 Rev: 1



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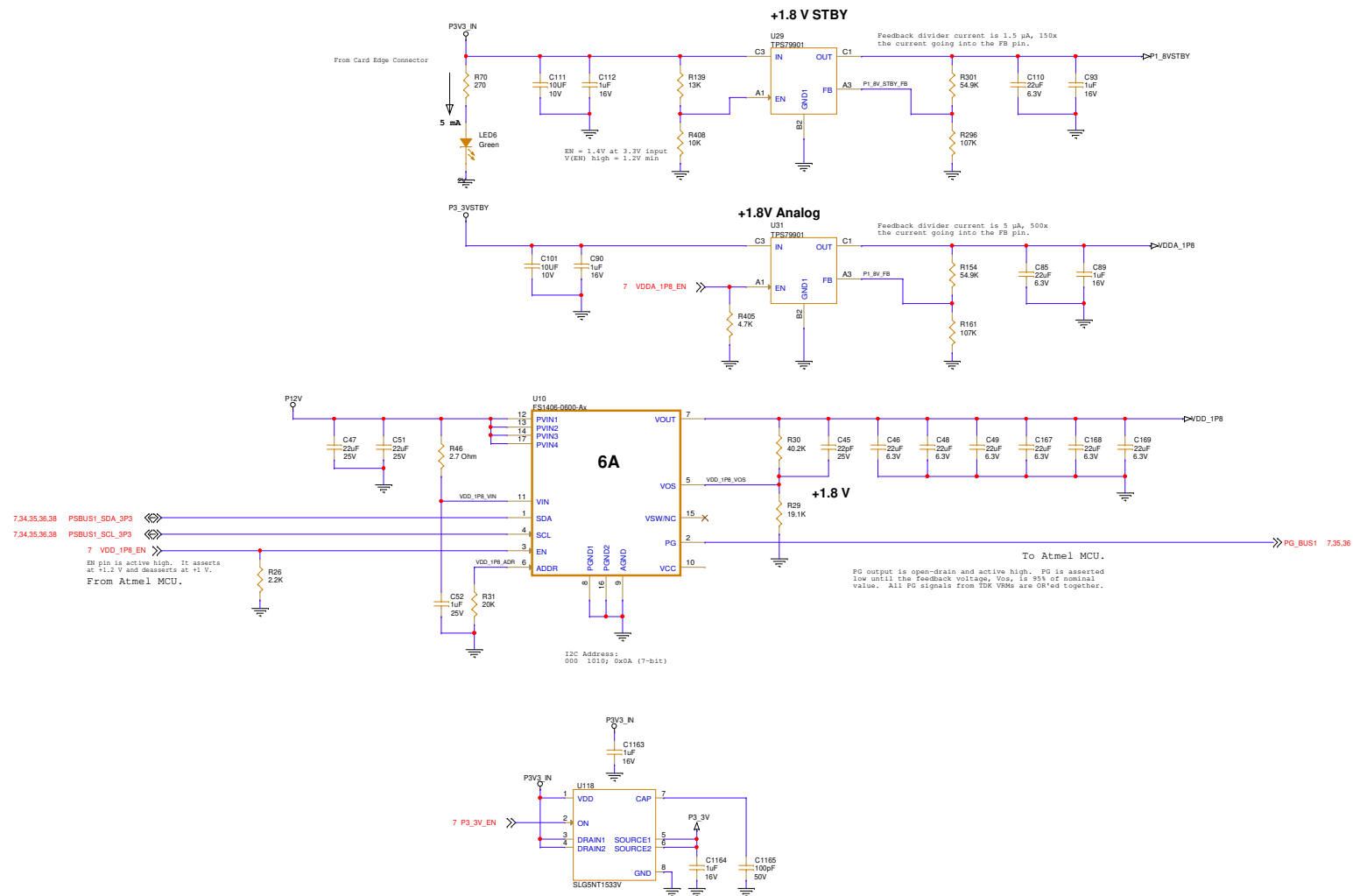
Doc Title: MXM	Doc Num: 400-00050
Page Title: 35 - PWR VDD QLP, MXN, LOGIC	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 35 of 38 Rev: 1



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Doc Title: MXM	Doc Num: 400-00050
Page Title: 36 - PWR VDD DDR Q 1P5	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 36 of 38 Rev: 1

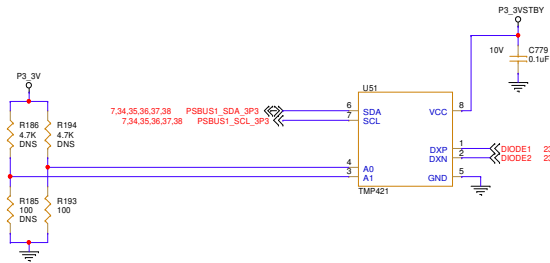


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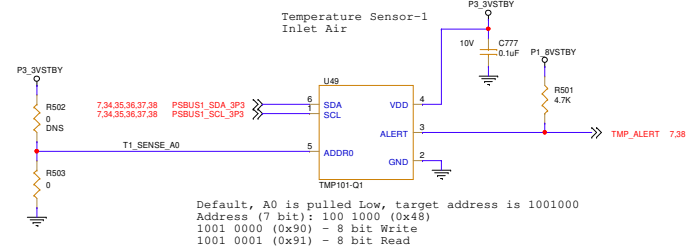


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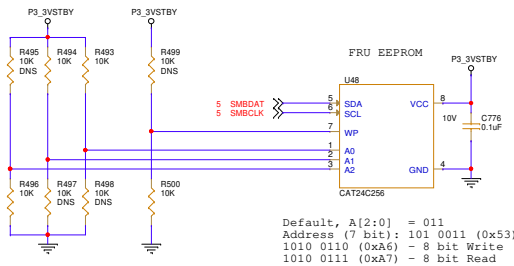
Doc Title: MXM	Doc Num: 400-00050
Page Title: 37 - PWR 1.8V, 3.3V Switched	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Size: C Sheet: 37 of 38 Rev: 1



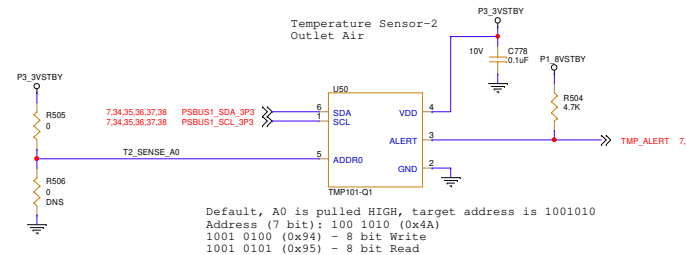
I2C Address:
 Default, A0 is pulled Low and A1 is float.
 001 1100; 0x1C (7-bit)
 0011 1000; 0x38 (8-bit write)
 0011 1001; 0x39 (8-bit read)



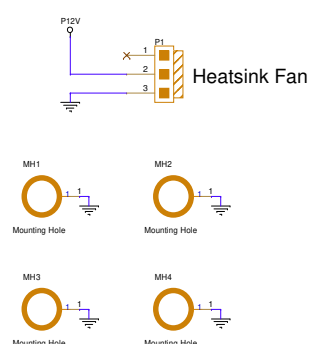
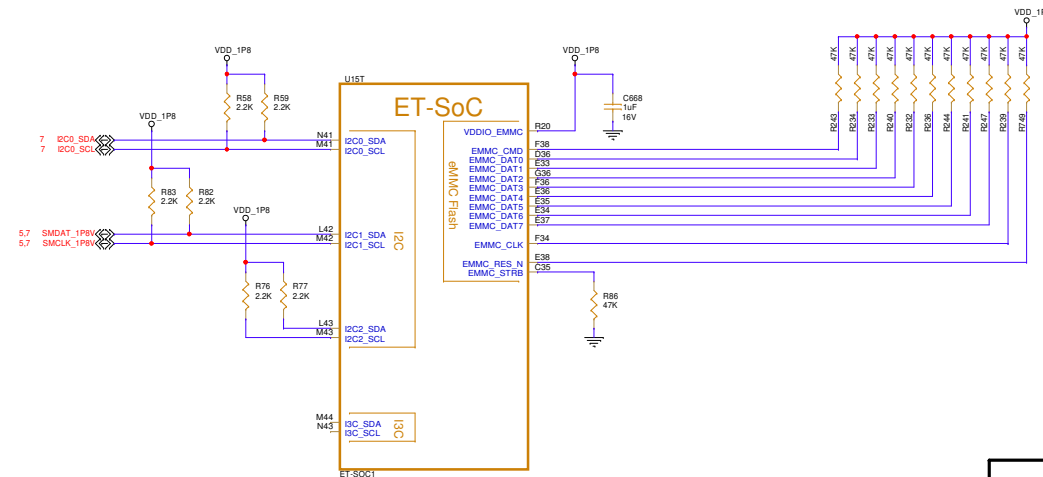
Default, A0 is pulled Low, target address is 1001000
 Address (7 bit): 100 1000 (0x48)
 1001 0000 (0x90) - 8 bit Write
 1001 0001 (0x91) - 8 bit Read



Default, A[2:0] = 011
 Address (7 bit): 101 0011 (0x53)
 1010 0110 (0xA6) - 8 bit Write
 1010 0111 (0xA7) - 8 bit Read



Default, A0 is pulled HIGH, target address is 1001010
 Address (7 bit): 100 1010 (0x4A)
 1001 0100 (0x94) - 8 bit Write
 1001 0101 (0x95) - 8 bit Read



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Doc Title: MXM	Doc Num: 400-00050
Page Title: 38 - Diode Temp FRU EEPROM	Drawn By: Bruce Klein
Date: Monday, March 16, 2026	Sheet: 38 of 38 Rev: 1